ICSA 2026 (author)

ICSA 2026 Submission 131

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Submission 131	
Title	Agentic Pattern Selection for Federated Learning Systems
Paper	(Dec 13, 10:47) (previous versions)
Track	Research
Author keywords	Architectural Patterns Federated Learning AI Agents Large Language Models
Abstract	Federated Learning (FL) is a distributed approach where multiple clients collaboratively train a global model without sharing their raw training data. Recent studies evaluate the adoption of architectural patterns in designing FL systems, as they can lead to both benefits and drawbacks, e.g., improved model accuracy and longer training times. Selecting architectural patterns is indeed not trivial, as their impact relies on the intrinsic characteristics of FL systems, such as clients' hardware resources and network conditions, as well as their runtime evolution, e.g., possible hardware failures and network degradation. This paper introduces an agentic approach to making architectural decisions at the end of each FL round, based on the metrics of interest (e.g., model accuracy and training time) collected so far, thereby determining which patterns are likely to improve these metrics in the next FL round. Agents rely on Large Language Models (LLMs) and operate by adopting three coordination patterns defined in the literature, specifically: Voting-based, Role-based, and Debate-based. Intuitively, the interaction among agents is regulated by three mechanisms: voting for a decision that is finalized by a coordinator, playing a role that is taken into account in the stage of converging towards a solution, and exchanging opinions and motivations, thus reaching an informed consensus. Results show that our approach, by adaptively selecting architectural patterns over FL rounds, achieves up to a 10% improvement in learning accuracy, and roughly 50% reduction in total execution time, at the cost of a moderate overhead.
Submitted	Nov 28, 14:52
Last update	Dec 07, 18:42
Artifact	https://zenodo.org/uploads/17838344 https:anonymous.4open.science/r/ICSA26-Prompts/README.md
Author conflicts	Luciano Baresi, Luciano Baresi, Alessio Bucaioni, Alessio Bucaioni, Tomas Bures, Matteo Camilli, Vittorio Cortellessa, Martina De Sanctis, Martina De Sanctis, Daniele Di Pompeo, Amleto Di Salle, J. Andres Diaz-Pace, Matteo Esposito, David Garlan, Jasmin Jahić, Patricia Lago, Qinghua Lu, Qinghua Lu, Ivano Malavolta, Raffaela Mirandola,

	Henry Muccini, Henry Muccini, Patrizio Pelliccione, Patrizio Pelliccione, Luciana Rebelo, Luciana Rebelo, Patrizia Scandurra, Alessandra Somma, Romina Spalazzese, Chouki Tibermacine, Catia Trubiani, Catia Trubiani, Roberto Verdecchia (view)

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Reviews

Review 1

Overall evaluation

-1: weak reject
Summary:
This paper presents an approach for adapting architectural patterns in Federated Learning systems to optimize inference accuracy and inference time. The core contribution consists of an agent-based solution where agents select one of three architectural patterns (Client Selection, Data Augmentation, Message Compression) at each training round to speed up inference and improve accuracy. The authors evaluate several agent coordination strategies involving vote-based, role-based, and debate-based cooperation. The experimental evaluation shows that the presented approach improves the accuracy and training time in Federated Learning systems over randomly selecting one of the three available strategies at each round.

Strengths:
+ The paper deals with an interesting and timely problem, and attemps to combine solutions for dealing with various problems that occur in Federated Learning.
+ The overall system architecture is well motivated and aligns with the goal of enabling runtime adaptation of FL architectural patterns.
+ The idea of using AI agents to dynamically adapt to changes is promising, especially that such agents can provide reasoning capabilities under different contexts.

Weaknesses:
- The paper lacks a concrete use case demonstrating why runtime pattern adaptation is necessary in Federated Learning.
- Discussion of Federated Learning literature is short.
- The experiments lack an evaluation against existing approaches, and omit many important details.

Comments for improving the paper:
- It's not very clear what dynamic changes may prompt changing the FL architectural pattern at runtime. I see that each pattern is designed to deal with a specific problem (e.g., resource heterogeneity, non-uniform data distribution), but a use case showing the need for adapting the pattern at runtime due to some changes would motivate the problem more clearly.
- The paper provides a good background about Federated Learning and the three architectural patterns considered. It's not clear however how data heterogeneity is detected

	if data is not shared with the central server. Is this a process that the client implements? Moreover, the computation overhead for data augmentation (e.g., using GANs) can sometimes be significant and not feasible on resource-constrained clients. - It's not clear from Figure 3 how architectural pattern decisions are propagate to the central server and client device. There is a link between the AI Agent Interface and the Architectural Pattern Manager, but I don't see any connection between the Architectural Pattern Manager and the components responsible for the architectural patterns (at the Client and Server side). It's also not clear how this is implemented in practice. - I don't see the need for using an LLM Coordinator Agent in the Voting-based strategy (Figure 4), since aggregating votes (majority or weighted) can be done with a simple rule-based component. This also applies to the role-based coordination strategy, where the coordinator agent only aggregates all recommendations and applies the resulting configuration. - The input for the agents described in Section III. C is not clearly defined: does it include information about the clients, performance metrics, etc.? - The authors emulate unstable network conditions by adding latency, which is not affected by the transmitted message sizes. It would be more appropriate here to limit bandwidth resources, which emulates network congestion more accurately. - It seems that the input to the AI agents are not very well defined. For example using terminologies like high-spec/low-spec clients, stable/unstable connections is ambiguous and depends largely on the context and the use case. Why haven't exact numbers been included in the system specifications? - I am not convinced that evaluating a random selection strategy is particularly meaningful in practice, as such a strategy would rarely be deployed in real systems. It would have been more appropriate to evaluate each pattern in isolation versus adapting patterns at runtime. Comparison to existing approaches (e.g., a rule-based approach) is also missing. - The changes to clients that are occuring at runtime are not clearly defined in the evaluation section. - Fig. 8b shows that the random strategy has a higher training time than other strategies. It's not clear whether this training time includes only the time during which the clients are performing model updates, or if it also includes the overhead incurred by selecting the correct strategy. Since Figure 9 shows that AI agents take around 50 seconds to reason about the strategy to select, wouldn't this add significant overhead to the training time? - It's unclear whether the (time) overhead resulting from data augmentation and message compression is taken into account. I would assume that message compression overhead can be negligible, but what about the data augmentation process? - Related work discussion of Federated Learning approaches is quite short and needs to be expanded, especially that there exists many papers that aim to mitigate resource heterogeneity/data heterogeneity and optimize accuracy and training time in Federated Learning settings.
Artifact availability	✔

Review 2	
Overall evaluation	**-1**: weak reject This paper proposes an approach, based on agentic AI, to make architectural decisions at the end of each Federated Learning (FL) round, based on the metrics of interest (e.g., model accuracy and training time), thus determining which architectural patterns are likely to improve these metrics in the next FL round. Agents rely and operate by adopting specific coordination and interaction patterns. While the topic is timely and relevant to the software architecture community, the paper does not clearly articulate its novelty or scientific contribution. In particular, it remains unclear what new architectural knowledge, methods, or insights are contributed beyond the instantiation and implementation of existing architectures and patterns.

Strengths
* The paper addresses an important and emerging problem: adaptive architectural decision-making in Federated Learning systems.
* The use of agentic AI to automate architectural decisions is promising and aligns with recent trends in self-adaptive and intelligent systems.
* The work is grounded in existing architectural knowledge, including reference architectures and coordination patterns, which gives it a good technical foundation.

Weaknesses
* The scientific novelty and contribution are not clearly defined, especially with respect to the software architecture community.
* The distinction between instantiating, extending, or proposing a new reference architecture is unclear.
* The paper lacks clarity on what architectural decisions are actually being made, despite this being central to the abstract and motivation.
* Some terminology and section titles are inconsistent or potentially misleading.
* The evaluation does not convincingly demonstrate how the proposed approach supports architects in making architectural decisions.

-- Detailed Comments --

Novelty and Contribution

The paper does not sufficiently explain its novelty. The proposed approach appears to instantiate an existing reference architecture and apply well-known coordination and interaction patterns among agents. As such, the contribution seems to lie primarily in the implementation of a specific solution rather than in advancing architectural theory, methods, or reusable knowledge.
In particular, it is unclear:
* What is new compared to existing approaches that combine FL with adaptive or agent-based systems.
* What architectural knowledge or reusable insights are contributed to the software architecture community.
The abstract promises “an approach to making architectural decisions at the end of each FL round,” but the paper does not clearly specify which architectural decisions are made (e.g., selection of patterns, reconfiguration options, component allocation). This aspect should be made explicit.

Terminology and Structure

The title of Section III is confusing. Methodology often is used as term to refer to the research methodology used in order to build the solution proposed in the work or more in general to perform the study. Also, be coherent in the use of the terms. The abstract describe the contribution of the work as “an agentic approach”. In this section, the contribution is called methodology.

Architectural Design Decisions and Motivation

Several design choices are insufficiently motivated. For example, the approach builds on the FLRA reference architecture, but the rationale for this choice is unclear:
* Is FLRA the only available reference architecture for FL systems?
* Does FLRA align well with the stated goal of optimizing performance metrics such as accuracy and training time?

	* Are there limitations of FLRA that motivate its selection or adaptation? These questions are particularly important given that the goal of the work is architectural optimization. The choice of reference architecture should be justified in this context. The authors state that they extend the FLRA architecture. However, it is not clear whether "extension" is the correct term: * What exactly is missing in FLRA that necessitates an extension? * Does the extension result in a new reference architecture? * Or is the work instead an instantiation or specialization of FLRA for a particular use case? Clarifying this point is essential to understanding the architectural contribution of the paper. Evaluation ---------- The evaluation section raises several concerns. The relationship between the three research questions (RQs) and the proposed approach is not sufficiently clear. A reader would expect the evaluation to assess: * How effective the approach is in making architectural decisions at the end of each FL round. * How well it supports software architects in their decision-making process. While RQ2 appears somewhat related to these aspects, the connection should be made more explicit. Referring to the conclusions, the evaluation should clearly demonstrate how and to what extent the approach supports architectural design decisions for FL systems, rather than focusing primarily on implementation-level or system-level performance aspects. Presentation - Minor comments: ----------------- * Figure 1 is too small. * Figure 7 appears very similar to Figure 1, with the main difference seemingly being Step 5. It is not clear whether both figures are necessary. The authors should consider: * Clearly explaining the distinct purpose of each figure, or * Merging them into a single, more informative figure to avoid redundancy. * be coherent in using "e.g.," vs "e.g.", and similar * avoid breaking "Figure 2a" (and similar) in two lines. Suggestions for Improvement --------------------------- * Clearly articulate the architectural novelty and contribution, explicitly positioning the work within the software architecture literature. * Specify which architectural decisions are made by the agents and how these decisions relate to established architectural concepts (e.g., patterns, tactics, trade-offs). * Clarify whether the work extends, instantiates, or derives a new reference architecture, and adjust terminology accordingly. * Improve consistency in terminology and section naming. * Strengthen the evaluation by aligning the research questions more directly with the stated goal of supporting architectural decision-making.
Artifact availability	✔

Review 3

Overall evaluation	**-2**: reject Summary & key contributions This paper reports a framework aimed at supporting the dynamic selection of patterns to improve the Federated Learning (FL) systems. The framework largely takes leverage two pieces of work: architectural design patterns for FL systems (ref#9) and Agent design patterns catalogue (ref#25); the reported work's framework introducing mutli-agents layer to enable architectural decisions of selecting the most suitable architectural patterns after each round of FL depending upon how an FL system is performing depending up a set of metrics. The Agentic AI components are being supported by LLM, for which purpose 3 well known LLMs have been used, e.g., Llama , Deepseek, and gpt-oss; the designed architecture of LLM+RAG is being supported by coordinator by three mechanisms: Voting-based, Role-based, and Debate-based. One of the key purpose of the reported work is to extend a FL system design framework called, FLRA – three main architectural patterns of FL used in this research are: Client Selector, Heterogeneous Data Handler and Message Compressor. The researchers have implemented the three mechanisms and a RAG system and run a set of experiments to answers three key questions aimed at finding out the most suitable LLM configuration is most suitable for supporting multiple agents layer; how effective the agentic layer is to improve FL systems, and the overhead of using multiple agent. The results have shown some improvement both accuracy and reduction in execution time. The authors claimed three key contributions as below: A framework for hosting the implementation of AI-Agents; Design of experiments to demonstrate the benefits of the proposed architectural framework; The flexibility in changing the design settings for patterns. Among these claimed contributions, the extension of FLRA by incorporating multiple agents appears to be a contribution worth considering. The other two contributions are dependents upon or required for demonstrating the first contribution – which in itself is not significant as a large part of the reported work is built based on the existing body of knowledge reported in ref#9 and ref#15 – it looks like that this work has been carried out to demonstrate the viability of work reported in ref#15 and the work in ref#9 is an application area, FL systems – for further details about this major concerns, please see comments under different heading below. Relevance The reported work is quite relevant to the software architecture discipline. It aims at bringing multi agents to support architectural decisions dynamically – every round of FL, agents considered the values of different metrics to improve the performance of the FL system. Soundness A large part of the work appears to be sound as it is based on pre-existing architectural patterns for designing FL systems and agentic AI; the authors have motivated the carried research, in this case to take advantage agentic ai for architectural decision during operations of a system. The authors have acknowledged the validity threats; however, there have not fully detailed the measures taken to minimize the potential negative impact of the identified validity threats. Moreover, the experimental set ups quite small and there is no specific justification for limiting 10 rounds of FL. The metrics used are for assessing FL – however, there is not much focus on the main aspects of the extension, assessing agentic AI and RAG supporting the architectural changes – Evals for an AI/Agentic systems are becoming quite critical part of any development/operations for a reasonable systems except a toy example. Given key architectural components are based on the existing work, it would be reasonable to expect sufficient attention paid to Evals for the multiple-agents in this system. Originality. The work is largely based on existing body of knowledge aimed at supporting pattern-driven architecture design for Federated Learning Systems and Agentic AI. The metrics used

	for evaluating the performance of FL system as a result of using LLM based Agents; indirectly assessing the performance of multiple agents systems – F. Given the work is aimed at extending an existing architectural framework for Federated learning, FLRA, it is expected that the reported work would be built upon the existing work, hence, it makes sense to provide detailed discussions on the relevant architectural patterns and coordination mechanisms. However, it is imperative on the authors to clearly and unambiguously describe the original contributions in a separate section to make it clear the reported work is more than implementing a few patterns/coordination mechanisms and running a set of experiments; the prompt structures used – zero short and a few short are reasonable but it looks too simplistic from scalability and reliability perspectives. The authors should produce a table to motivation and justify the research questions as there is only one research questions, #2, whose answer can determine the viability of the produced solutions – a contribution. Overall, it is not clear to me how much contributions this work makes to the design and operations of architecture of FL systems. This point should be thoroughly addressed. Comments on other parts of the paper. Discussion There should be in-depth discussion on the findings with broader applications rather than further discussing the architectural strategies like role-based, voting-based, debate-based. It is clear that overhead of agents would be expected; hence, the findings of minor improvement in accuracy and overhead increased could be expected. The assessment of the introduced components (directly for LLM/agents) should be direct rather than through proxy. Quality – the paper is well written but too verbose; the authors have spent too much space on explaining different theoretical and architectural concepts that are well known and widely understood in architecture research and practice – design patterns for FL systems, archiectures for agentic AI systems. Related section – it creates a lot of repetition as most of the literature mentioned in this section has already been described and discussed a lot in the first few sections of the papers. And the related section doesn't really add any value in appreciating the theoretical foundations, technical implementations and the key finding. I would argue to remove it and use that space for produced insightful discussions and reflections.
Artifact availability	✔

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